

AMA HIRUSHIMA

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ABOUT ME

Instrumentation and Automation Technology graduate with a passion for embedded systems, automation and IoT. Experienced in working with microcontrollers, sensors, data acquisition systems, and industrial automation technologies through academic projects and practical training. Seeking opportunities to apply and expand my technical skills while contributing to engineering and technology-driven projects.

WORK EXPERIENCE

Trainee Embedded Systems Engineer (Internship) February 2025 - July 2025
Sri Lanka Telecom - Digital Lab

Mathematics Tutor for (G.C.E) Ordinary Level Students (Part-Time)

EDUCATION

Bachelor of Engineering Technology (Hons) in Instrumentation and Automation
Faculty of Technology,
University of Colombo.

G.C.E (Advanced Level)Technology Stream - 2020 (Mahinda Rajapaksha College, Homagama)

Certificate Courses

PLC Programming - (SL Institute of Robotics) - SLIR
Successfully completed a PLC Programming course with practical training in ladder logic, HMI/SCADA, industrial networking, motor control, and automation systems using Siemens, Mitsubishi, and Omron PLC platforms.

PROJECTS

VLC-Based Passive-Powered Sensing and Control Module for Indoor Farming **2026**
(Final Year Research Project)
Developed a final year research project focused on a VLC-based embedded control system capable of reading multiple sensor inputs and controlling multiple actuators. The system uses Visible Light Communication (VLC) with IR uplink for data transmission and a downlink control node for bidirectional communication. The module operates using light-based energy harvesting, eliminating the need for wired power supply. Edge computing techniques are implemented to enable on-device data processing, intelligent decision-making, and efficient handling of sensor data for real-time control and automation

Automated 3D Printing Platform **2024**
Developed a web-based 3D printing service platform with automated pricing and designer matching, using React, Flask, Node.js, and MongoDB, integrated with PrusaSlicer for workflow automation. - Group Project

Temperature Monitoring System **2024**

Developed a Temperature Monitoring System using the Atmega328P microcontroller. Technologies Used: AVR Programming, Interrupts, 4-digit 7-segment display - Group Project

Green House Monitoring System **2024**

Developed a LabVIEW-based greenhouse monitoring system to acquire and visualize environmental data, using potentiometers to simulate NPK levels, along with sensors for light intensity and CO_2 monitoring.

Automated Soil Moisture Monitoring and Watering System **2022**

Developed an Automated Soil Moisture Monitoring and Watering System Arduino UNO-based system.

SKILLS

Technical Skills

- ProgrammingLanguages: C,C++,Python,HTML,CSS,VHDL,Octave,Ladder Logic
- Microcontrollers: ESP32, Arduino, MSP430,ATmega328P
- Firmware & IDEs: ESP-IDF, Arduino IDE, code composer studio,
- VS Code Instrumentation / DAQ: LabVIEW
- 3D CAD: SolidWorks

Soft Skills

strong problem-solving, teamwork, and communication skills developed through academic projects and practical training. Quick to learn new technologies and committed to delivering high-quality work with attention to detail.

Communication & Documentation: Clear technical documentation and teamwork in project-based environments.

ADDITIONAL INFORMATION

Languages

Sinhala: Native
English: Advanced

REFERENCES

Mr. Eng. Rumesh Rajasegar
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Sri Lanka Telecom PLC
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